ANNEXURE-1

Standard Template for Academic Tasks (26271)

Academic Task Number: 2 **Course code: CSE570**
Date of allotment:8 Sep 2026 **Course title: MACHINE LEARNING WITH PYTHON**
Date of submission: 8 NOV 2026 **Maximum Marks: 100**
Academic Task Type: Skill Based Assignment **Mode: Individual - Online**

Question Number	Question Statement	Course Outcome	Bloom's level	Marks per Question
Q1	**A) Problem Statement (Objectives) and Dataset - 10 Marks** **Project Theme:** End-to-End Machine Learning Solution for a Real-World Problem **1. Problem Statement - 5 Marks:** Select a meaningful classification or regression problem. Clearly define the problem context, target variable, machine-learning task, measurable objectives/success criteria, intended users or impact, and key assumptions/constraints. **2. Dataset - 5 Marks:** Select a suitable public or authorized dataset (e.g., Kaggle, UCI, government/open-data portal or equivalent). Provide the source link/citation, number of records and features, target variable, feature types, data dictionary, class/target distribution, missing-value summary, and justification for why the dataset is appropriate for the stated problem.	CO1	L4: Analyze	10
Q2	**B) Implementation (Outcome), Report, and Viva - 60 Marks** **1. Implementation - 30 Marks** **a) Data Cleaning and Visualization - 5 Marks:** Load the data in Python and perform required cleaning such as duplicate handling, missing-value treatment, datatype correction, outlier inspection/treatment, encoding, scaling/normalization where applicable, and meaningful visualizations that reveal important data characteristics. **b) EDA and Statistical Analysis - 10 Marks:** Carry out univariate, bivariate and multivariate exploratory analysis. Use appropriate descriptive statistics, correlation/association analysis, distributions, group-wise	CO2, CO3	L6: Create	60

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	comparisons and at least one suitable statistical test or confidence-based analysis wherever relevant. Clearly interpret the findings rather than only displaying plots/tables. **c) Model Development and Evaluation - 15 Marks:** Create a baseline and develop at least three appropriate machine-learning models in Python. Use a proper train/validation/test strategy or cross-validation, perform required preprocessing without data leakage, tune important hyperparameters, compare models using problem-appropriate metrics, and justify the final model. For classification, use suitable measures such as precision, recall, F1-score, ROC-AUC and confusion matrix; for regression, use suitable measures such as MAE, RMSE and R^2. Include interpretation of model behaviour/important features and identify limitations. **2. Report - 10 Marks** **a) Format - 5 Marks:** Submit a professionally organized report containing title page, problem statement, objectives, dataset description, methodology/workflow, EDA, model development, results, model comparison, conclusion, limitations, references, and links to GitHub/LinkedIn evidence. **b) Technical Writing - 5 Marks:** Use clear technical language, correct terminology, labelled figures/tables, concise result interpretation, consistent formatting, and proper citation of datasets, libraries, external material and references. **3. Viva - 20 Marks:** Each student must individually explain the complete project, justify preprocessing and model choices, interpret key plots and metrics, demonstrate important parts of the Python implementation, answer conceptual questions, and explain limitations/improvement possibilities. Inability to explain submitted work may reduce viva marks.			
Q3	**C) LinkedIn Engagement - 10 Marks** Publish one professional LinkedIn post summarizing the problem, dataset,	CO4	L5: Evaluate	10

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	approach, one important visual/insight, best-model result, key learning, and GitHub repository link. Submit the post URL and a dated screenshot as evidence. **1. Likes - 6 Marks** **2. Comments - 4 Marks**			
Q4	**D) GitHub Contributions - 20 Marks** Maintain an individual GitHub repository containing a clear README, problem/objectives, dataset source, Python notebook/scripts, requirements/dependencies, outputs/results and report. Use incremental version control and submit the repository URL. **1. Commits - 10 Marks:** Meaningful, progressive commits with clear messages reflecting actual development stages. **2. Pull Requests - 5 Marks:** Use branch-based development and pull requests for significant changes/features with meaningful titles/descriptions. **3. Stars - 5 Marks:** Verified repository stars available by the submission deadline.	CO4	L3: Apply	20

Important Note:

1. In case, any faculty members wants to split any Question into two sub parts of 5 marks each the same can be done after the consent from the concerned HOD.
2. In case of MCQ based CA or term paper or design problem etc. the faculty member can add or remove the number of rows as required.
3. The template for academic task is only applicable for theory courses and not for practical and Project/Dissertation courses.
4. Each faculty member must also upload the soft copy of the duly filled template on the UMS interface at the time of CA allocation.
5. The faculty can also use this standard template as Question paper after adding the fields such as Reg.No, Roll No, Section, Group, Set Number etc.
6. In case any academic task is mapped with two course outcomes then it is necessary that two Questions out of three (one each for a course outcome CO) must be as per the particular course outcome whereas the third question can be from framed from any of the two course outcomes for that particular academic task as per choice of the faculty member.
7. For more detailed information regarding the outcome based evaluation and revised bloom's taxonomy, kindly refer to Annexure-II: Examination reform policy